

TASK 2: Deploy a Virtual Machine & Install Web

ServerObjective:

The goal of this task is to launch an EC2 instance on AWS, configure the network firewall, and install an Apache web server to display a custom webpage.

Technical Specifications:

Instance OS: Ubuntu 24.04 LTS.

Instance Type: t2.micro (Free Tier Eligible).

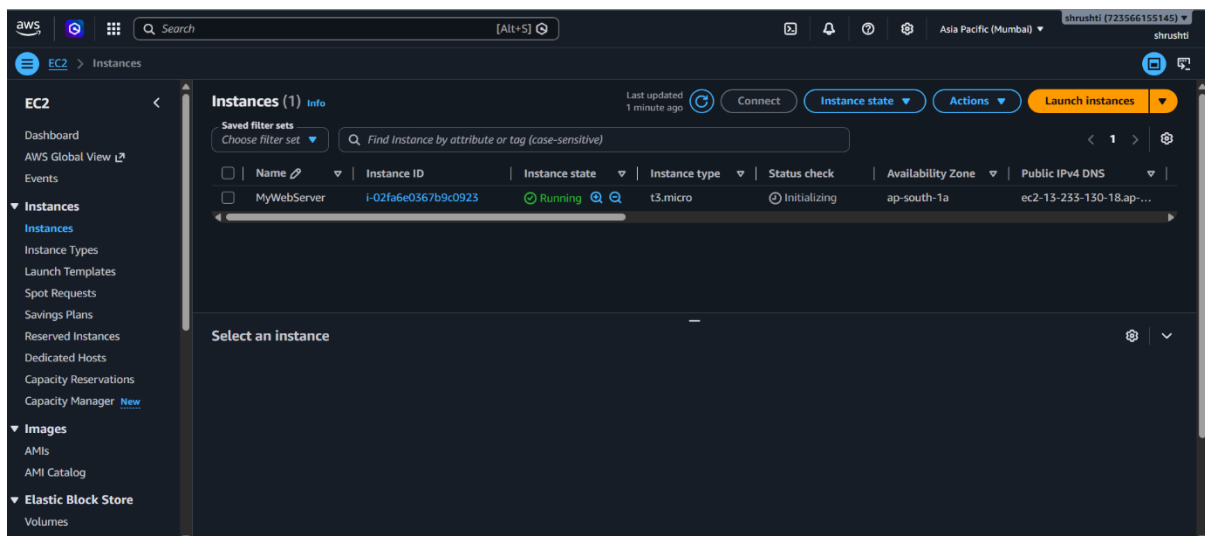
Firewall Rules: Security Group configured to allow traffic on Port 80 (HTTP) and Port 22 (SSH).

Software: Apache2 Web Server

Step 1: Launching the EC2 Instance

Logged into the AWS Management Console and navigated to the EC2 service.

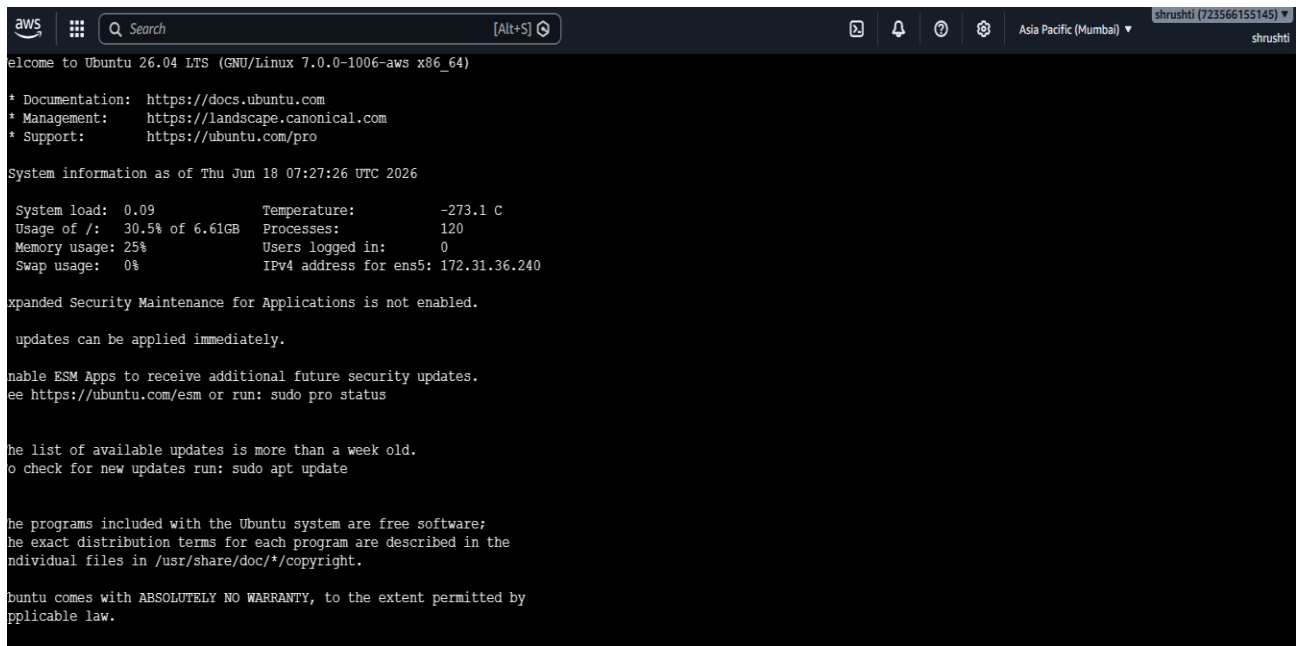
initiated the "Launch Instance" process, named the instance "MyWebServer," and selected "Ubuntu" as the operating system



Step 2: Configuring Security Group

Accessed the network settings and edited the Inbound Rules.

Added rules for HTTP (Port 80) and SSH (Port 22) to enable public access to the web server.



```
aws [Q Search] [Alt+S] Asia Pacific (Mumbai) shrushti (723566155145) shrushti
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu Jun 18 07:27:26 UTC 2026

System load:  0.09           Temperature:   -273.1 C
Usage of /:    30.5% of 6.61GB Processes:      120
Memory usage: 25%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.36.240

Expanded Security Maintenance for Applications is not enabled.

Updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Step 3: Installing Apache Web Server

Connected to the instance via the terminal.

Executed system updates and installed the Apache web server

Commands: `sudo apt update`

`sudo apt install apache2 -y`

`sudo systemctl start apache2`

```
aws [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) shrushti (723566155145) shrushti
nabling conf localized-error-pages.
nabling conf other-vhosts-access-log.
nabling conf security.
nabling conf serve-cgi-bin.
nabling site 000-default.
reated symlink /etc/systemd/system/multi-user.target.wants/apache2.service' -> '/usr/lib/systemd/system/apache2.service'.
reated symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' -> '/usr/lib/systemd/system/apache-htcacheclean.service'.
rocessing triggers for ufw (0.36.2-9build1) ...
rocessing triggers for man-db (2.13.1-1build1) ...
rocessing triggers for libc-bin (2.43-2ubuntu2) ...
canning processes...
canning linux images...

unning kernel seems to be up-to-date.
o services need to be restarted.
o containers need to be restarted.
o user sessions are running outdated binaries.

o VM guests are running outdated hypervisor (qemu) binaries on this host.
buntu@ip-172-31-36-240:~$ sudo systemctl start apache2
buntu@ip-172-31-36-240:~$ echo "<h1>Hello from Shrushti's Cloud VM</h1>" | sudo tee /var/www/html/index.html
^C
buntu@ip-172-31-36-240:~$ ^C
buntu@ip-172-31-36-240:~$ echo "<h1>Hello from Shrushti's Cloud VM</h1>" | sudo tee /var/www/html/index.html
^C
buntu@ip-172-31-36-240:~$ sudo rm /var/www/html/index.html
buntu@ip-172-31-36-240:~$ echo "<h1>Hello from Shrushti's Cloud VM</h1>" | sudo tee /var/www/html/index.html
h1>Hello from Shrushti's Cloud VM</h1>
buntu@ip-172-31-36-240:~$
```

Step 4: Final Output

Accessed the default Apache page by entering the instance's Public IP address into a web browser.

Deliverable: Public IP: [3.26.48.209].

